Mei Luo, Hailong Peng, Zhonghua Deng, Zhonghua Yin, Qiang Zhao and Hua Xiong*

Preparation and Characterization of Genipin-Crosslinked Chitosan Microspheres for the Sustained Release of Salidroside

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Abstract: Chitosan microspheres (CsMs) that encapsulate salidroside (Sal) were prepared by the emulsion crosslinking method with naturally occurring genipin (Gp) and then examined for their *in vitro* release. Sal-loaded CsMs (Sal-CsMs) showed nearly spherical and smooth surfaces with internal voids. The particle size of Sal-CsMs ranged within 0.56–5.01 μm, and their encapsulation efficiency and loading capacity were beyond 77.58% and 23.29%, respectively. The stability of Sal improved after entrapment into the CsMs. The release rate of Sal from CsMs was initially rapid, followed by sustained release. The release behavior depended on the pH of the release medium. The main release mechanisms underlying the release procedure were anomalous behavior and Fickian diffusion. These results indicated that CsMs with a novel crosslinker of Gp was a potential carrier system for producing functional foods containing Sal.

Keywords: salidroside, chitoson microsphere, genipin, crosslinking, sustained release

***Corresponding author: Hua Xiong,** State Key Laboratory of Food Science and Technology, Nanchang University, No. 235 Nanjing East Road, Nanchang 330047, Jiangxi, China, E-mail: huaxiong100@126.com
Mei Luo, Hailong Peng, State Key Laboratory of Food Science and Technology, Nanchang University, No. 235 Nanjing East Road, Nanchang 330047, Jiangxi, China; Department of Chemical Engineering, Nanchang University, No. 999 University Avenue, Nanchang 330031, Jiangxi, China, E-mail: luomei@ncu.edu.cn, peng_hailong@aliyun.com
Zhonghua Deng, State Key Laboratory of Food Science and Technology, Nanchang University, No. 235 Nanjing East Road, Nanchang 330047, Jiangxi, China, E-mail: hello_world_value@126.com
Zhonghua Yin, Department of Chemical Engineering, Nanchang University, No. 999 University Avenue, Nanchang 330031, Jiangxi, China, E-mail: m15170002676@163.com
Qiang Zhao, State Key Laboratory of Food Science and Technology, Nanchang University, No. 235 Nanjing East Road, Nanchang 330047, Jiangxi, China, E-mail: qiangzhao1221@yahoo.com

1 Introduction

Salidroside (Sal, p-hydroxyphenethyl-β-D-glucoside, Figure 1(A)), which is extracted from *Rhodiola rosea* L., is a bioactive component possessing various functional properties, such as anti-aging [1], antifatigue [2], antiviral [3], antihypoxia [4], anticancer [5], cardioprotection [6], and strong antioxidative activities [7]. Sal has been widely applied in the food and medical industries in the form of oral solution, capsule, and powder. However, the conventional forms of Sal for food processing or pharmaceutics have significant drawbacks because of their sensitivity to light and air, concentration fluctuation in blood, low bioavailability, and short biological half-life [8–10]. To overcome these drawbacks, researchers have developed technology for encapsulation of Sal, such as liposomes [10, 11] and silica nanoparticles [12]. Unfortunately, traditional and polymeric liposomes are unstable *in vivo*/*in vitro* and in complex synthesis processes, respectively, whereas silica nanoparticles are unsuitable for functional foods. Thus, constructing a natural polymer-based microsphere delivery system to encapsulate Sal may be an effective approach to overcome the aforementioned problems.

Natural polymer-based microsphere delivery systems, including gelatin, zein, and chitosan microsphere (CsMs), have gained tremendous interest in recent years because of their nontoxicity, good biocompatibility, and biodegradability [13–19]. However, gelatin microspheres often suffer from poor mechanical properties, easy dissolution in aqueous environments, and poor controlled release profile [13–15]. Additionally, the application of zein is restricted by its laborious extraction process from corn [16]. Cs is obtained from the deacetylation of chitin, which is second in abundance to cellulose. Compared with many other natural polymers, Cs is a cationic polysaccharide containing a number of free amine groups that are readily available for crosslinking. It has mucoadhesive character, which increases residual time at the site of absorption. Therefore, Cs has been widely investigated for its potential in delivery systems and food industry applications [18–23].

Different methods, such as ionotropic gelation, coacervation, emulsion crosslinking, solvent evaporation, and

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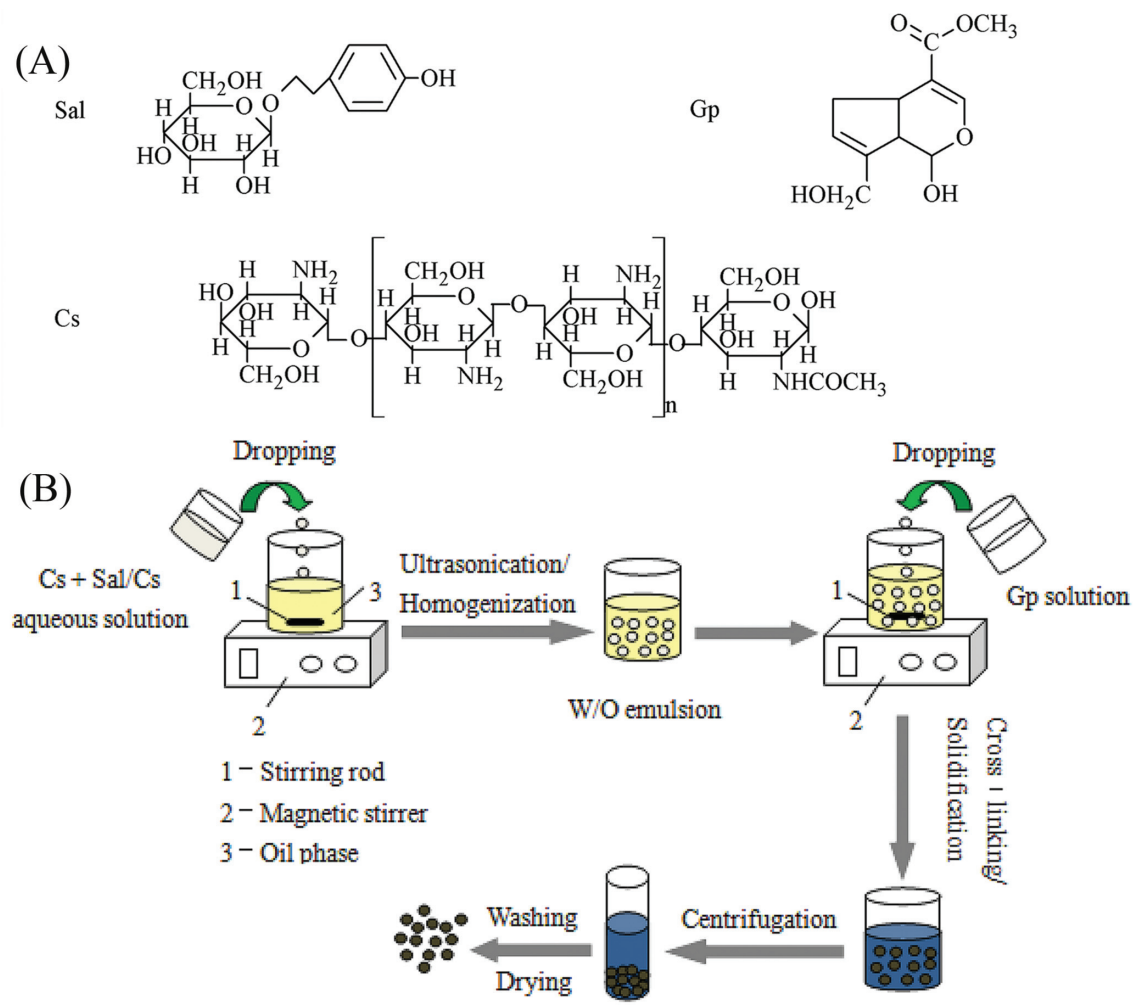

Figure 1: Schematic of the preparation of CsMs by emulsion crosslinking method: structures of Sal, Gp, Cs (A) and preparation process (B).

spray drying, have been used to prepare CsMs [24, 25]. Among these methods, the emulsion crosslinking method has been extensively employed for narrow particle size distribution and good morphology of the microspheres with prolonged retention time and controlled release of the loaded active agents [24, 26, 27]. Unfortunately, the most commonly used crosslinkers in emulsion crosslinking are chemically synthesized, such as glutaraldehyde, formaldehyde, tripolyphosphate, ethylene glycol, diglycidyl ether, and disulfide. All these chemical crosslinking reagents have physiological toxicities. Therefore, searching for natural and nontoxic crosslinkers for CsMs is highly necessary. Genipin (Gp, structure shown in Figure 1(A)), a fully biocompatible reagent approximately 5,000 to 10,000 times less cytotoxic than glutaraldehyde, is a natural water-soluble crosslinker isolated from the fruits of *Gardenia jasminoides* Ellis, and it is generally used as a component of traditional Chinese medicine [28, 29]. It reacts with biopolymers containing primary amine groups and changes them to dark blue [30]. Drug-loaded CsMs have been successfully obtained by crosslinking with Gp, which has better biocompatibility and slower degradation rates [19, 31]. However, to date, no investigation has been reported about Gp-crosslinked CsMs for loading of functional foods.

Thus, this study aimed to develop Gp-crosslinked CsMs as a delivery system for the loading and sustained release of Sal via the emulsion crosslinking method. The physicochemical characteristics, loading capacities, and *in vitro* release properties of the resulting Gp-crosslinked CsMs were characterized. Meanwhile, two possible influencing factors of Cs:Sal mass ratios and Gp ratios (wt% of Cs) on Sal release profiles were investigated. Furthermore, Peppas' model was used to investigate the probable release mechanisms of Gp-crosslinked CsMs.

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2 Materials and methods

2.1 Materials

Chitosan (Cs, molecular weight $= 2 \times 10^5$, degree of deacetylation $\geq$ 85%) was purchased from Gracia Chemical Technology Co., Ltd. (Chengdu, China). Salidroside (Sal, > 98.00%, wt%) was obtained from the National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). Genipin (Gp, > 98.00%, wt%) was supplied by Conbon BioTech Co., Ltd. (Chengdu, China). All other chemicals and reagents used were analytical grade.

2.2 Preparation of CsMs

Figure 1(B) is a schematic of the preparation of CsMs by emulsion crosslinking method. Cs (180 mg) was dissolved in 6 mL of 0.5 M acetic acid aqueous solution containing Tween-20 (2.5%, w/v). Then, a specific amount of Sal was directly mixed with the Cs solution under agitation at room temperature until bubbles disappeared, and this mixture was used as the water phase. Meanwhile, a mixture of liquid paraffin (light, 80 mL) and Span-80 (2%, w/v) was used as the oil phase. The water phase was dispersed dropwise onto the oil phase using an injector with stirring for 30 min at room temperature. The solution was continuously homogenized by ultrasonication for 5 min to form a W/O emulsion. Gp dissolved in 70% alcohol as the crosslinker was slowly added into the emulsion to solidify the Cs droplets for 3 h with agitation. The microparticle suspension was subsequently centrifuged at 4,000 rpm for 10 min. The Sal-loaded CsMs (Sal-CsMs) were then collected and sequentially washed with petroleum ether, acetone, and isopropyl alcohol, respectively. Finally, the samples were freeze dried and stored in a desiccator until further analysis. Sal-CsMs with different Cs:Sal mass ratios (1:1, 2:1, and 3:1) and different Gp ratios (10%, 20%, and 30%) were prepared. The %yield of Sal-CsMs was calculated as follows:

$$\text{Yield } (\%) = \frac{\text{mass of Sal-CsMs}}{\text{mass of feed Sal} + \text{Cs}} \times 100\% \qquad (1)$$

The plain CsMs (P-CsMs) were also prepared under identical conditions but without Sal in the reaction system.

2.3 Characterization

The morphological structure of microspheres was investigated under a scanning electron microscopy (SEM) system (Hitachi S-4800FE, Hitachi, Ltd., Tokyo, Japan). The microspheres were first fractured by a razor blade to investigate the inner structure. The samples were sprinkled onto a brass stub using double-sided adhesive carbon tape and then sputter-coated with gold under vacuum. The average particle size of the microspheres was measured from several SEM images by counting a minimum of 100 particles using Image-Pro Plus software (version 6.0) and Origin software (version 8.0). Fourier transform infrared spectroscopy (FTIR) of Gp, Cs, Sal, P-CsMs, Sal-CsMs, and P-CsMs/Sal physical mixture were recorded using an FTIR spectrophotometer (Nicolet 5700, Thermo Electron Corporation, Waltham, MA, USA). These spectra were collected from 4,000 cm^{-1} to 400 cm^{-1} at a resolution of 2 cm^{-1}. The tested samples were prepared by the KBr-disk method. Thermal stability analyses of Cs, Sal, P-CsMs, Sal-CsMs, and P-CsMs/Sal physical mixture were performed under a dynamic nitrogen atmosphere with a flow rate of 50 mL min^{-1} using a TG/DTA analyzer (Diamond TG/DTA, PerkinElmer Corporation, Waltham, MA, USA). TGA/DTA data were recorded at a heating rate of 10°C min^{-1} from 25°C to 500°C. X-ray diffraction (XRD) patterns of Gp, Cs, Sal, P-CsMs, Sal-CsMs, and P-CsMs/Sal physical mixture were obtained using a powder X-ray diffractometer (Empyrean, PANalytical B. V., Almelo, Netherlands) with a scanning speed of 4° min^{-1} in the 2θ range of 5–50° at ambient temperature. Samples were analyzed using a copper anode operated at a voltage of 40 kV with a current of 40 mV.

2.4 Sal entrapment of CsMs

Sal-CsMs (25 mg) were extracted using 0.1 M HCl aqueous solution (80 mL) for 24 h at 37°C under vigorous stirring. The solution was then metered the volume to 100 mL with HCl aqueous solution and filtered through 0.45 μm Millipore filters. The filtrate was analyzed by HPLC on a 1200 Series HPLC system (Agilent Technologies Co., Ltd., Santa Clara, CA, USA) at 275 nm wavelength. The chromatographic column was a Kromasil C_{18} column (4.6 mm × 250 mm; 5 μm). The mobile phase was $V_{\text{methanol}}/V_{\text{water}} = 15/85$ at a flow rate of 1.0 mL min^{-1}, and the injection volume was 10 μL at 30°C. The Sal loading capacity (LC) and encapsulation efficiency (EE) were calculated according to the following formulas:

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$$LC_{actual}\,(\%) = \frac{\text{mass of Sal loaded in microspheres}}{\text{mass of Sal-CsMs}} \times 100\% \quad (2)$$

$$LC_{theoretical}\,(\%) = \frac{\text{mass of feed Sal}}{\text{mass of feed Sal} + \text{Cs}} \times 100\% \quad (3)$$

$$EE\,(\%) = \frac{LC_{actual}}{LC_{theoretical}} \times 100\% \quad (4)$$

2.5 Stability

Sal powders (reference samples) and Sal-CsMs were stored in a weighing bottle for 10 d under the following conditions: (1) high temperature (60°C) in darkness, (2) high humidity (relative humidity = 92.5%), and (3) light irradiation (4500 ± 500 lx). The related properties were assessed at different sampling times (0, 5, and 10 days). Changes in LC (%) of Sal-CsMs and the loss in mass of pure Sal (%) were measured, respectively, to evaluate the effect of embedding on Sal stability in temperature and light treatments, respectively. Moreover, stability of both samples in humidity treatment was evaluated by the increase in weight (g) of microspheres with an initial weight of 0.05 g.

2.6 *In vitro* release

In vitro release experiments for Sal-CsMs were performed in pH 1.2 diluted hydrochloric acid (HCl) and pH 7.4 phosphate-buffered saline (PBS). Sal-CsMs (10 mg) were dispersed in 50 mL screw-capped glass tubes filled with 40 mL of release media at 37 ± 0.5°C. The suspension was gently shaken at 100 rpm using a horizontal shaking water bath. At predetermined time intervals, the tubes were centrifuged at 4,500 rpm for 5 min, and 3 mL of supernatant was withdrawn using a Millipore filter (0.45 μm pore size). Meanwhile, an equal volume of the corresponding fresh medium warmed at 37 ± 0.5°C was added to the tubes to maintain a constant volume, and the suspension was subsequently redispersed. The amount of Sal in the filtered samples was determined by HPLC. The cumulative release of Sal was calculated from the ratio of the cumulative mass of Sal released from microspheres and total loading amount of Sal in the microspheres. Release data were fitted according to Peppas' model:

$$\ln M_t/M_\infty = n\ln t + \ln k \quad (5)$$

where M_t is the mass of Sal released at time t, M_∞ is the total Sal (i.e., M_t/M_∞ is the fractional release of Sal at time t), k is a constant incorporating the properties, and n is the diffusional exponent indicating the release mechanism.

All experiments were conducted by three times, and the results are expressed as mean ± standard deviation.

3 Results and discussion

3.1 Formation of CsMs

Stable water-in-oil (w/o) emulsion could be easily prepared by emulsifying the Cs aqueous solution in the oil phase. When Gp was added to the Cs/liquid paraffin emulsion, the great mass of Gp may be partitioned into the continuous phase. Therefore, the crosslinking reaction between Cs and Gp was slow and led to gradual solidification of microspheres from the Cs emulsion droplets, which was conducive to the good sphericity of microspheres. Figure 2 shows the crosslinking of Cs with Gp. The reaction mechanism is as follows [19, 31]:

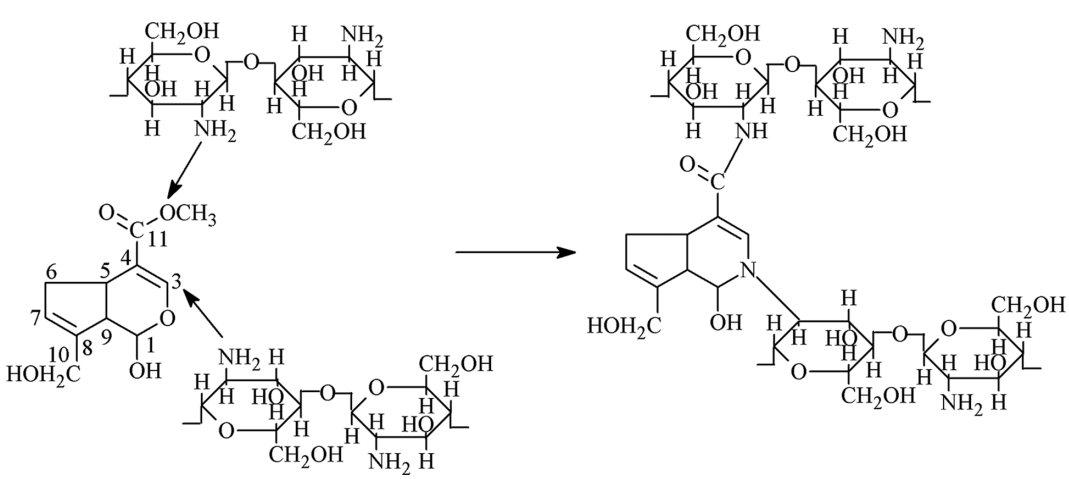

Figure 2: Crosslinking reaction between Cs and Gp.

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(1) the olefinic carbon atom at C-3 of Gp is subjected to nucleophilic attack by the amino group of Cs, and the dihydropyran ring is opened to form heterocyclic amine; (2) the ester group of Gp is nucleophilically substituted by the amino group on Cs to form amide.

The yield (%) of Sal-CsMs was found to range from 55.64 ± 1.58% to 82.84 ± 2.02% with varying Cs:Sal and Gp ratios. As shown in Table 1, the yield (%) was found to be proportional to the Cs:Sal and Gp ratios. These results may be attributed to the increase in EE with an increase in the Cs:Sal ratio, as well as more fully crosslinking between Cs and Gp with an increase in the Gp ratio.

Table 1: Yield, average size, and Sal loading of Sal-CsMs.

Sal-CsMs			Yield (%)	Average size (μm)	LC_{actual} (%)	EF (%)
Code	Cs:Sal mass ratio	Gp ratio (wt. % of Cs)				
Sal-CsMs_1	1:1	20	75.33 ± 2.33	2. 07 ± 1.31	44.34 ± 0.78	88.69 ± 1.56
Sal-CsMs_2	2:1	20	79.60 ± 2.54	2.59 ± 0.80	30.11 ± 0.43	90.43 ± 1.32
Sal-CsMs_3	3:1	20	81.83 ± 1.65	4.14 ± 1.17	23.29 ± 0.31	92.34 ± 1.23
Sal-CsMs_4	2:1	10	55.64 ± 1.58	4.14 ± 1.17	25.83 ± 1.01	77.58 ± 3.03
Sal-CsMs_5	2:1	30	82.84 ± 2.02	2.84 ± 0.93	28.12 ± 0.59	84.46 ± 1.78

Note: All values are mean ± standard deviation ($N = 3$).

3.2 Physicochemical characteristics of CsMs

The morphology and size of CsMs were investigated by SEM, and the typical results are shown in Figure 3. P-CsMs (Figure 3(A) and 3(B)) and Sal-CsMs (Figure 3(E) and 3(F)) were found to have nearly spherical, fairly smooth, and uniform surfaces. Some irregular small particles existed on the spherical surface (Figure 3(B) and 3(F)), which could be attributed to the effects of mechanical stress during stirring or of the movement of moisture during drying [26]. The microsphere matrices were also investigated through their broken surfaces (Figure 3(C) and 3(G)). The crosslinked microsphere matrices were found to have compacted and continuous networks with numerous voids. The void structure may be related to the formation of air bubbles or diffusion of water from emulsion droplets to the continuous phase during crosslinking and solidification. The typical size distribution of CsMs is shown in Figure 3(D) and 3(H). The microspheres were relatively homogeneous in size, ranging from 0.56 μm to 5.01 μm in diameter. The average particle size of Sal-CsMs was slightly larger than that of P-CsMs. This finding may be due to the increased solution viscosity with the addition of functional Sal components. The average particle sizes of all the Sal-CsMs are given in Table 1. The results showed that the size of Sal-CsMs increased with increasing Cs:Sal mass ratio, which may be ascribed to the increased solution viscosity. However, the Gp ratio did not exert any perceptible effect on particle size. These results were in agreement with the findings of Kawadkar and Chauhan [19].

The FTIR spectra of Gp, Cs, P-CsMs, Sal-CsMs, and P-CsMs/Sal mixture are shown in Figure 4(A). In the FTIR spectrum of Gp, the characteristic peaks at 3,404, 3,243, and 1,682 cm^{-1} were assigned to the two hydroxyl groups and C = O group of carboxylic ester, respectively. The stronger absorption bands at 1,622 and 880 cm^{-1} were attributed to C = C stretching vibration and = C–H bending vibration of the heterocyclic ring, respectively. The stretching vibrations of different –C–H groups corresponded to the peaks at 2,943–2,391 cm^{-1}. As shown in the FTIR spectrum of Cs, an absorption band at 3,446 cm^{-1} was attributed to the overlapped stretching vibration of –OH and N–H, and characteristic peaks of the saccharide structure around 891 and 1,156 cm^{-1} appeared. Two characteristic peaks at 1,656 and 1,599 cm^{-1} were assigned to C = O stretching vibration (amide I) and N–H bending vibration (including amide II), respectively. The absorption of C–H stretching of methyl or methylene group of Cs was attributed to the peak at 2,920 cm^{-1}. Meanwhile, the peak at 1,381 cm^{-1} corresponded to the C–H deformation vibration of the methyl group. Lastly, the absorption bands at 1,098 and 1,026 cm^{-1} were attributed to the stretching vibrations of C–O on Cs. For P-CsMs, Sal-CsMs, and P-CsMs/Sal mixture, the peak of –OH/N–H of Cs shifted from 3,446 cm^{-1} to 3,416 cm^{-1}, which may be due to the increased π bonds of the Schiff base (C = N). A weak hydrogen bond association was found between the hydrogen of –OH and the π electron cloud of C = N. The significantly increased

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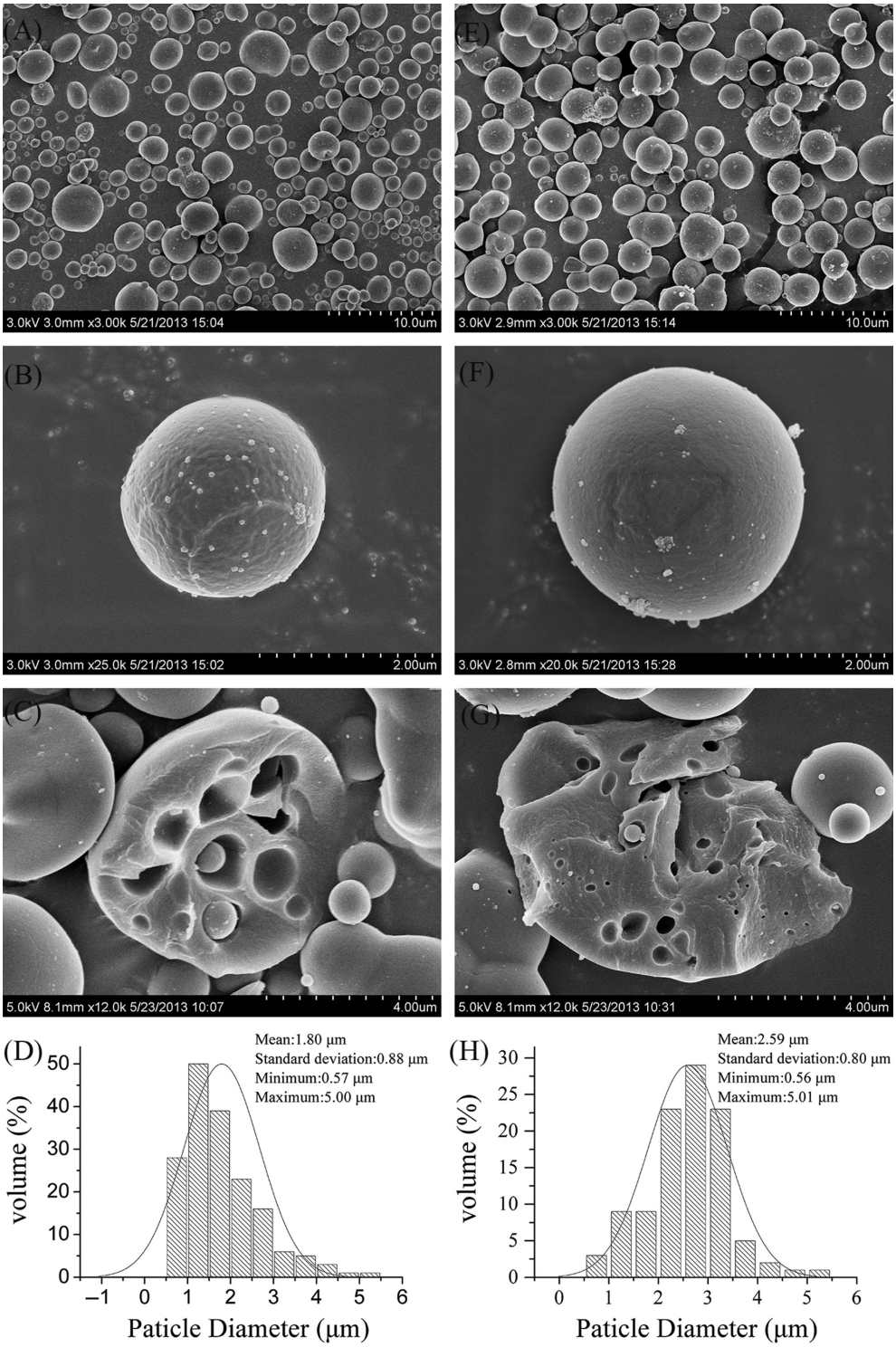

Figure 3: Typical morphological features of P-CsMs (A~D) and Sal-CsMs (F~H): surface morphology (A, B, E, F), internal structure (C, G), particle size distribution (D, H).

adsorption at 1,622 cm^{-1} was attributed to the formation of a secondary amide group between the amino groups on Cs and the ester groups of Gp. Meanwhile, the decreased adsorption at 1,599 cm^{-1} was ascribed to nucleophilic substitution by the amino group of Cs on the olefinic carbon atom at C–3 of Gp, followed by the opening of the dihydropyran ring to form a heterocyclic amine [19, 31]. These results indicated that Cs was

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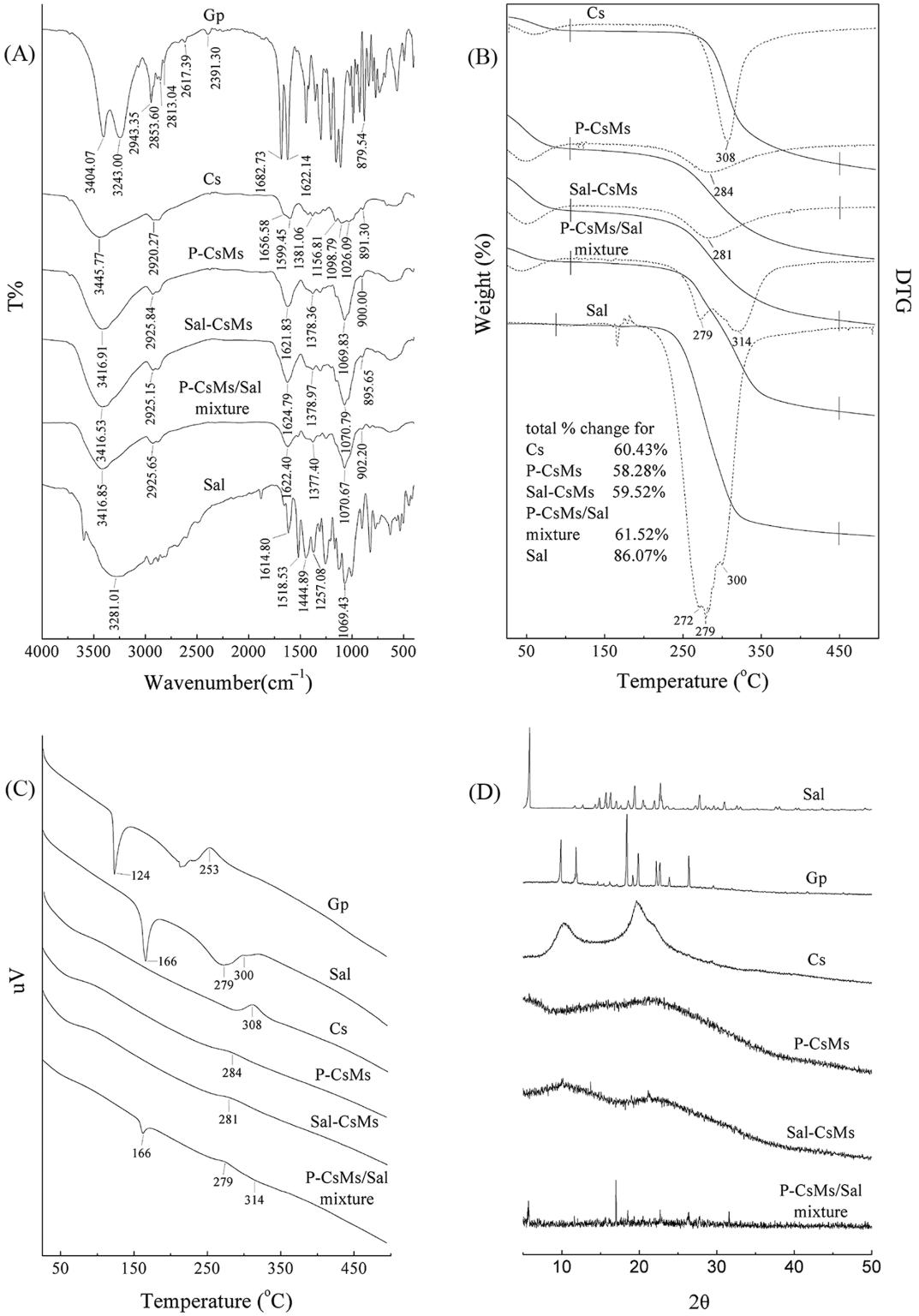

Figure 4: FTIR spectra (A), TGA thermograms (B), DTA thermograms (C) and X–ray diffractograms (D).

successfully crosslinked by Gp through Schiff base reaction and nucleophilic substitution. The FTIR characteristic peaks of Sal at 3,281, 1,257, and 1,069 cm^{-1} corresponded to the stretching vibrations of –OH, C–O–C, and C–O groups, respectively. Meanwhile, the characteristic stretching vibration absorption of benzene ring corresponded to the three peaks at 1,615, 1,519, and 1,445 cm^{-1}, respectively. In the two spectra of Sal-CsMs

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and P-CsMs/Sal mixture, the characteristic peaks of Cs along with Sal all appeared to have no significant difference from those of P-CsMs. These results indicated that no interaction occurred between Cs and Sal.

Results of TGA and DTA analyses of Cs, P-CsMs, Sal-CsMs, P-CsMs/Sal mixture, and Sal are presented in Figure 4(B) and 4(C). Weight loss below 100°C was attributed to water or the low-boiling-point solvents evaporating from the samples. For the Cs powder, weight loss was constant from 120°C to 260°C, and an abrupt weight loss was observed when above 260°C. A narrow degradation temperature range with a maximum weight loss rate at 308°C, which corresponded to the exothermic peak of Cs decomposition at 308°C (Figure 4(C)), was also found. By contrast, Sal showed a broader degradation temperature range, in which weight loss occurred in three superimposed stages with the peaks at 272°C, 279°C, and 300°C, respectively. P-CsMs and Sal-CsMs had broad degradation patterns with maximum weight loss rates at 284°C and 281°C, respectively. The conformation change in Cs that resulted from Gp crosslinking may influence the formation of new phases of the polymer, which partly decreases bond energies of glycosidic linkages in Cs, consequently leading to its thermal instability compared with Cs powder [32]. The difference in temperature from 284°C down to 281°C may be due to the decomposition of entrapped Sal. Moreover, P-CsMs/Sal mixture also had broad degradation patterns and showed two weight loss stages with the peaks at 279°C and 314°C, respectively, which corresponded to the weight loss superposition of P-CsMs and Sal. The second stage with a lagging peak at 314°C may be ascribed to the strong hydrogen bonding interaction between Sal and the decomposition products of P-CsMs, such as acetic acid, butyric acid, and a series of lower fatty acids [33]. As shown in Figure 4(C), pure Sal showed a sharp endothermic peak at 166°C corresponding to the melting point of Sal in the crystalline form. The same peak was observed at reduced intensity in the P-CsMs/Sal mixture, indicating the presence of crystalline Sal outside the microspheres. By contrast, the absence of the endothermic peak in the thermogram of Sal-CsMs was ascribed to the amorphous or molecularly dispersed structure of Sal in the Cs matrix as a solid solution [34], which was further supported by XRD studies.

The XRD patterns of Sal, Gp, Cs, P-CsMs, Sal-CsMs, and P-CsMs/Sal mixture are shown in Figure 4(D). In the diffraction patterns of Sal and Gp, a series of sharp peaks at different angles indicated that Sal and Gp had crystalline properties. Characteristic XRD patterns of Cs were observed at 2θ of 10° and 20°. The diffraction patterns were also observed at reduced intensity at the same point as that of Sal in the P-CsMs/Sal mixture, suggesting that Sal was not completely dispersed. On the contrary, the diffraction patterns of P-CsMs and Sal-CsMs showed no marked peak, indicating that the microspheres were in an amorphous state, as supported by DTA studies. These results also clearly suggested that Cs and Gp were successfully crosslinked, and Sal was encapsulated in the microspheres.

3.3 Sal entrapment

The EE and LC of Sal in microspheres were evaluated by HPLC, and the typical HPLC graphs of Sal are shown in Figure S1. Sal entrapment of all the Sal-CsMs is also shown in Table 1. The EE of Sal in the Sal-CsMs was found to be from 77.58 ± 3.03% to 92.34 ± 1.23%. The EE increased with increasing Cs:Sal ratio, which may be attributed to the increase viscosity of Cs solution. This increase prevented Sal from leaving the internal phase. However, LC decreased with increasing EE. These results agreed with those reported by Dhawan and Singla [35]. Meanwhile, the Gp ratios did not show any pronounced effects on Sal entrapment.

3.4 Stability

According to the values of yield, EF, LC, and morphology, Sal-CsMs crosslinked with 20% Gp at 2:1 Cs:Sal ratio (Sal-CsMs_2) were selected for stability investigation. Sal easily degrades under light, heat, and humidity conditions, thereby restricting its application in the functional food industry. Entrapping Sal into CsMs could effectively hinder its exposure to adverse environments and improve its stability. The stability of Sal-CsMs and Sal under different conditions was investigated, and the results are shown in Table 2. The content of pure Sal significantly decreased by approximately 5.00% within 10 d under high temperature and light irradiation, whereas the decrease of Sal in CsMs was nearly 1.00%. In addition, the weight of pure Sal increased much more significantly than that of CsMs under high humidity. These results may be due to the protective effect on the internal Sal caused by the physical incorporation of Sal into the spherical polymer matrix.

3.5 *In vitro* release and release kinetics

The Sal release profiles were plotted between cumulative release of Sal and release time. The effects of Sal loading content, Gp ratio, and release medium on Sal release

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Table 2: Stability of the CsMs and pure Sal.

Samples	Sampling time (d)	Type of treatment		
		Temperature	Light	Humidity
Sal-CsMs	0	30.11 ± 0.43[a]	30.11 ± 0.43[a]	0.0500 ± 0.0004[c]
	5	29.53 ± 0.21[a]	29.44 ± 0.25[a]	0.0518 ± 0.0011[c]
	10	29.08 ± 0.23[a]	29.04 ± 0.17[a]	0.0524 ± 0.0010[c]
Pure Sal	0	99.11 ± 0.51[b]	99.11 ± 0.51[b]	0.0500 ± 0.0003[c]
	5	96.32 ± 0.67[b]	97.30 ± 0.28[b]	0.0596 ± 0.0046[c]
	10	94.02 ± 0.55[b]	94.68 ± 0.49[b]	0.0654 ± 0.0051[c]

Notes: [a]Stability of the Sal-CsMs samples was evaluated by changes in LC (%) in temperature and light treatments. [b]Stability of pure Sal samples was evaluated by the loss in mass of powders (%) in temperature and light treatments. [c]Stability of both samples in humidity treatment was evaluated by the increase in weight (g) of microspheres with an initial weight of 0.05 g. All values are mean ± standard deviation ($N = 3$).

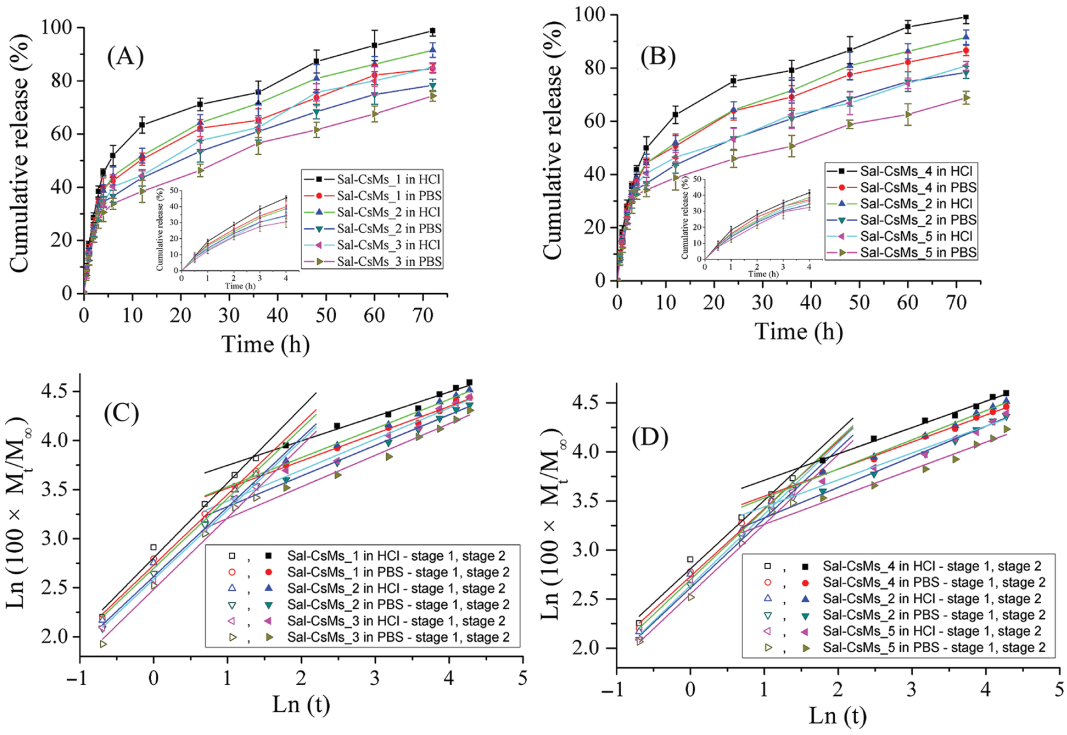

Figure 5: *In vitro* release profiles (A, B) and release kinetics (C, D) of Sal from CsMs in pH 1.2 HCl and pH 7.4 PBS solutions. Values are mean ± standard deviation ($N = 3$).

profiles from Sal-CsMs are presented in Figure 5(A) and 5(B). Figure 5(A) shows that the released Sal from CsMs with higher Cs:Sal ratio could be controlled more easily compared with Sal-CsMs with lower Cs:Sal ratio, which may be due to increasing density of polymer matrix and an increase in the diffusional path length that the Sal molecules have to transverse [19]. The results of *in vitro* Sal release for the formulations Sal-CsMs_2, Sal-CsMs_4, and Sal-CsMs_5 are shown in Figure 5(B). Total cumulative release of Sal at 72 h decreased from 99.24 to 80.84 and from 86.56 to 68.83 in HCl (pH 1.2) and PBS (pH 7.4) solutions, respectively, when the Gp ratio increased from 10% to 30%. When crosslinking increased, the rigidity of the polymer matrix increased, which allowed slower diffusion and erosion compared with the polymer matrix crosslinked at low Gp ratio. As shown in Figure 5(A) and 5(B), the release of Sal from CsMs in HCl solution exhibited a higher rate than that in PBS solution for the same feed Cs:Sal ratio or Gp ratio, which may be attributed to the protonation of Cs amino groups and fast swelling in acidic medium. Moreover, the release of Sal from CsMs comprised two stages. Sal had a burst effect in

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the early stage of release in both media, which may be due to the rapid diffusion of Sal from the surface of CsMs. Subsequently, the release slowed down at an almost constant rate. The burst release helped rapidly reach the effective concentration of Sal, whereas the sustained release maintained the effective concentration of Sal in plasma. These results indicated that the delivery system of CsMs was conducive to controlling the Sal release profiles by the encapsulation method and prolonging the biological half-life by sustained release.

Table 3: Release exponent n value and regression coefficient (R^2) for Sal releasing from CsMs.

Code	Model	HCl solution		PBS solution	
		Stage 1	Stage 2	Stage 1	Stage 2
Sal-CsMs_1	n	0.7646	0.2489	0.7198	0.2778
	R^2	0.9851	0.9729	0.9921	0.9848
Sal-CsMs_2	n	0.7125	0.2973	0.7073	0.3123
	R^2	0.9975	0.9936	0.9929	0.9928
Sal-CsMs_3	n	0.7061	0.3161	0.7323	0.3214
	R^2	0.9968	0.9661	0.9872	0.9718
Sal-CsMs_4	n	0.6977	0.2681	0.6888	0.2757
	R^2	0.9830	0.9895	0.9862	0.9914
Sal-CsMs_5	n	0.7275	0.2744	0.7101	0.2791
	R^2	0.9919	0.9688	0.9924	0.9659

The Sal-release kinetics of CsMs was characterized by fitting the experimental data to Peppas' model [36]. The exponent n value and regression coefficient R^2 were calculated (Table 3), and the results are shown in Figure 5(C) and 5(D). The exponent n values for stage 1 were found to be from 0.6977 to 0.7646 and from 0.6888 to 0.7323 in HCl (pH 1.2) and PBS (pH 7.4) solutions, respectively. These results suggested that atypical Fickian diffusion ($0.43 < n < 0.85$) was the major release mechanism for stage 1, which could be attributed to the dissolution of surface-adhered Sal on the microspheres. Additionally, the calculated n values for stage 2 were from 0.2489 to 0.3161 and from 0.2757 to 0.3214 in HCl (pH 1.2) and PBS (pH 7.4) solutions, respectively. This result indicated that Fickian diffusion ($n < 0.43$) was the major release mechanism for stage 2 and Sal release from CsMs was driven by the concentration gradient of Sal probably resulting from the increased hydrophobicity of the crosslinked CsMs.

4 Conclusion

Sal-CsMs were successfully prepared for the first time using the emulsion crosslinking method. Gp, an herbal medicine approximately 5,000 to 10,000 times less cytotoxic than glutaraldehyde, was used as a crosslinking agent. The crosslinking mechanism was Schiff base reaction and nucleophilic substitution. The CsMs had smooth surfaces and continuous networks. Sal stability improved after being entrapped into CsMs. The release of Sal from CsMs in the dissolution medium was initially rapid, after which an obvious sustained process was observed. The Sal-release kinetics of CsMs was characterized by Peppas' model. Therefore, Gp-crosslinked CsMs may have potential application as a delivery system of functional foods.

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References

1. Mao G, Deng H, Yuan L, Li D, Wang Z. Protective role of salidroside against aging in a mouse model induced by D-galactose. Biomed Environ Sci 2010;23:161–6.
2. Ma L, Cai D, Li H, Tong B, Song L, Wang Y. Anti-fatigue effects of salidroside in mice. J Med C PLA 2008;23:88–93.
3. Wang H, Ding Y, Zhou J, Sun X, Wang S. The in vitro and in vivo antiviral effects of salidroside from *Rhodiola rosea* L. Against coxsackievirus B3. Phytomedicine 2009;16:146–55.
4. Li QY, Wang HM, Wang ZQ, Ma JF, Ding JQ, Chen SD. Salidroside attenuates hypoxia-induced abnormal processing of amyloid precursor protein by decreasing BACE1 expression in SH-SY5Y cells. Neurosci Lett 2010;481:154–8.
5. Sun C, Wang Z, Zheng Q, Zhang H. Salidroside inhibits migration and invasion of human fibrosarcoma HT1080 cells. Phytomedicine 2012;19:355–63.
6. Zhang J, Liu A, Hou R, Zhang J, Jia X, Jiang W, et al. Salidroside protects cardiomyocyte against hypoxia-induced death: a HIF-1α-activated and VEGF-mediated pathway. Eur J Pharmacol 2009;607:6–14.
7. Gupta V, Lahiri SS, Sultana S, Tulsawani RK, Kumar R. Antioxidative effect of *Rhodiola imbricata* root extract in rats during cold, hypoxia and restraint (C-H-R) exposure and post-stress recovery. Food Chem Toxicol 2010;48:1019–25.
8. Yu S, Liu L, Wen T, Liu Y, Wang D, He Y, et al. Development and validation of a liquid chromatographic/electrospray ionization mass spectrometric method for the determination of salidroside in rat plasma: application to the pharmacokinetics study. J Chromatogr B 2008;861:10–15.

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9. Wei Q, Ge T, Kong S. Salidroside promotes erythropoiesis and protects erythroblasts against oxidative stress by up-regulating glutathione peroxidase and thioredoxin. J Ethnopharmacol 2011;133:308–14.
10. Fan M, Xu S, Xia S, Zhang X. Effect of different preparation methods on physicochemical properties of salidroside liposomes. J Agr Food Chem 2007;55:3089–95.
11. Peng H, Li W, Ning F, Yao L, Luo M, Zhu X, et al. Amphiphilic chitosan derivatives-based liposomes: synthesis, development, and properties as a carrier for sustained release of salidroside. J Agr Food Chem 2014;62:626–33.
12. Peng H, Dong R, Wang S, Zhang Z, Luo M, Bai C, et al. A pH-responsive nano-carrier with mesoporous silica nanoparticles cores and poly(acrylic acid) shell-layers: fabrication, characterization and properties for controlled release of salidroside. Int J Pharm 2013;446:153–9.
13. Tuan Hiep T, Ramasamy T, Poudel BK, Marasini N, Bo Kyung M, Hyuk Jun C, et al. Preparation and characterization of spray-dried gelatin microspheres encapsulating ganciclovir. Macromol Res 2014;22:124–30.
14. Jin YC, Kim IY, Kim ID, Lee HK, Park JY, Han PL, et al. Biodegradable gelatin microspheres enhance the neuroprotective potency of osteopontin via quick and sustained release in the post-ischemic brain. Acta Biomater 2014;10:3126–35.
15. Lan Y, Li WC, Guo R, Zhang Y, Xue W, Zhang YM. Preparation and characterisation of vancomycin-impregnated gelatin microspheres/silk fibroin scaffold. J Biomater Sci-Polym Ed 2014;25:75–87.
16. Rishi S, Zein MC. The industrial protein from corn. Ind Crop Prod 2001;13:171–92.
17. Liu X, Sun Q, Wang H, Zhang L, Wang JY. Microspheres of corn protein, zein, for an ivermectin drug delivery system. Biomaterials 2005;26:109–15.
18. Jameela SR, Kumary TV, Lal AV, Jayakrishnan A. Progesterone-loaded chitosan microspheres: a long acting biodegradable controlled delivery system. J Control Release 1998;52:17–24.
19. Kawadkar J, Chauhan MK. Intra-articular delivery of genipin cross-linked chitosan microspheres of flurbiprofen: preparation, characterization, in vitro and in vivo studies. Eur J Pharm Biopharm 2012;81:563–72.
20. Agnihotri SA, Aminabhavi TM. Novel interpenetrating network chitosan-poly(ethylene oxide-g-acrylamide) hydrogel microspheres for the controlled release of capecitabine. Int J Pharm 2006;324:103–15.
21. Casettari L, Vllasaliu D, Lam JKW, Soliman M, Illum L. Biomedical applications of amino acid-modified chitosans: a review. Biomaterials 2012;33:7565–83.
22. Belscak-Cvitanovic A, Stojanovic R, Manojlovic V, Komes D, Cindric IJ, Nedovic V, et al. Encapsulation of polyphenolic antioxidants from medicinal plant extracts in alginate-chitosan system enhanced with ascorbic acid by electrostatic extrusion. Food Res Int 2011;44:1094–101.
23. Srinivasa PC, Baskaran R, Ramesh MN, Prashanth KVH, Tharanathan RN. Storage studies of mango packed using biodegradable chitosan film. Eur Food Res Technol 2002;215:504–8.
24. Agnihotri SA, Mallikarjuna NN, Aminabhavi TM. Recent advances on chitosan-based micro- and nanoparticles in drug delivery. J Control Release 2004;100:5–28.
25. Sinha VR, Singla AK, Wadhawan S, Kaushik R, Kumria R, Bansal K, et al. Chitosan microspheres as a potential carrier for drugs. Int J Pharm 2004;274:1–33.
26. Peng H, Xiong H, Li J, Xie M, Liu Y, Bai C, et al. Vanillin cross-linked chitosan microspheres for controlled release of resveratrol. Food Chem 2010;121:23–8.
27. Zhang Y, Yu Y, Shi X, Zhao S, Chen A, Huang D, et al. Study on the preparation of genipin crosslinked chitosan microspheres of resveratrol and in vitro release. J Polym Res 2013;20:175–84.
28. Huang LH, Sung H, Tsai C, Huang D. Biocompatibility study of a biological tissue fixed with a naturally occurring crosslinking reagent. J Biomed Mater Res 1998;42:568–76.
29. Nickerson MT, Patel J, Heyd DV, Rousseau D, Paulson AT. Kinetic and mechanistic considerations in the gelation of genipin-crosslinked gelatin. Int J Biol Macromol 2006;39:298–302.
30. Butler MF, NG YF, Pudney PDA. Mechanism and kinetics of the crosslinking reaction between biopolymers containing primary amine groups and genipin. J Polym Sci A Polym Chem 2003;41:3941–53.
31. Mi F, Sung H, Shyu S. Release of indomethacin from a novel chitosan microsphere prepared by a naturally occurring cross-linker: examination of crosslinking and polycation-anionic drug interaction. J Appl Polym Sci 2001;81:1700–11.
32. He J, Lu YC, Luo GS. Ca(II) imprinted chitosan microspheres: an effective and green adsorbent for the removal of cu(II), cd(II) and pb(II) from aqueous solutions. Chem Eng J 2014;244:202–8.
33. Nieto JM, Penichecovas C, Padron G. Characterization of chitosan by pyrolysis-mass spectrometry, thermal analysis and differential scanning calorimetry. Thermochim Acta 1991;176:63–8.
34. Umadevi SK, Thiruganesh R, Suresh S, Reddy KB. Formulation and evaluation of chitosan microspheres of aceclofenac for colon-targeted drug delivery. Biopharm Drug Dispos 2010;31:407–27.
35. Dhawan S, Singla A. Nifedipine loaded chitosan microspheres prepared by emulsification phase-separation. Biotech Histochem 2003;78:243–54.
36. Ritger PL, Peppas NA. A simple equation for the description of solute release II. Fickian and anomalous release from swellable devices. J Control Release 1987;5:37–42.

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